

CHAPTER 1

TRAFFIC SOURCE MODELS

1.1 Introduction

The Information Flow model captures the average traffic generated by a source, but it overlooks the finer temporal dynamics. While this model is useful for rough network resource dimensioning or assessing system stability, it falls short when analyzing more specific performance metrics such as delay, traffic peaks, or for accurately sizing certain network components, such as buffers. For these purposes, it is necessary to describe the offered traffic profile, i.e., the pattern with which data are actually generated by the source, through suitable *traffic models*.

According to the final design goal, a traffic model can actually capture different aspects of the traffic generated by a source. For example, to have an idea of the order of magnitude of the bitrate required from a trunk connection, it suffices to model just the long-term average traffic offered by all the data sources multiplexed over that trunk. Instead, to check whether the average transmission delay of a certain Information Flow is below a given threshold, the model also needs to account for the time dynamics of the traffic source. In both cases, the actual content of the traffic is irrelevant for the network design and dimensioning, and hence it is usually not considered in traffic modeling. If the purpose is to design access schemes to support traffic prioritization, instead, the source model should also differentiate the packets generated by the source according to their priority class.

In the rest of this chapter, we will first illustrate a general method to describe a traffic source by means of mathematical models. Successively, we will discuss some basic traffic source models, which can be used to check the performance of some network management

protocols (Medium Access Control (MAC) and scheduling algorithms, in particular), or to provide a first, rough dimensioning of the network resources. Finally, we will present some methodologies to verify the model validity.

1.2 Abstractions of an Information Flow

An Information Flow is typically associated to a logical *session* between two (or more) End Nodes that need to exchange information. For example, when a web browser contacts a web server to get a certain World-Wide Web (WWW) content, the two End Nodes establish an Hypertext Transfer Protocol (HTTP) session. During the session, the nodes will actively exchange data in certain time intervals, and remain idle in others (e.g., when the user consults the content of the downloaded webpage). The session ends when the client connects to a different web server or it is closed by the user. A different example of a logical session between two End Nodes may be that of a mailer fetching new emails from a mail server. In this case, the session starts with the request from the client to the mail server, and ends when all the emails are downloaded to the mailer application. Therefore, traffic exchange during a session may be irregular, with alternating periods of active data exchange with periods of inactivity. During the active periods, furthermore, the traffic exchanged by the End Node can be constant or variable, according to the type of application.

Depending on the purpose of the traffic model, therefore, we can describe an Information Flow at different levels of abstractions, as depicted in Fig. 1.1, i.e.,

- *Session level*: the model only describes when an Information Flow starts and ends, and what is the average generated traffic. In other words, the model describes a logical *session* between two End Nodes;
- *Activity level*: the model describes the active and idle periods during a session, by defining the duration of such periods (usually, in statistical terms) and the average traffic generated in each period;
- *Packet level*: the model describes in details the packet generation pattern and packet size distribution during active and idle periods.

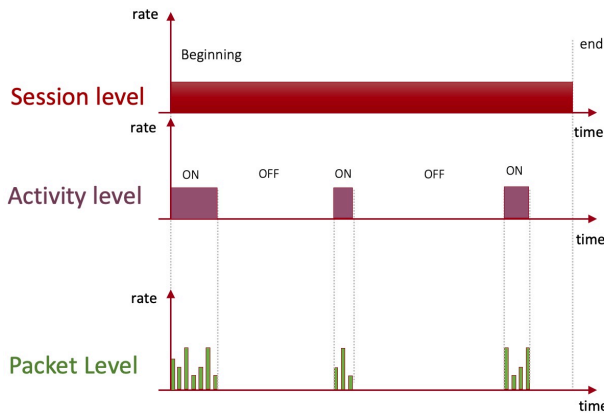


Figure 1.1 Graphical representation of the different levels of abstraction of a source model.

In the following, we first introduce a very general model that can be used to represent a large variety of traffic sources. Then, we will present some very common traffic source models that are often used in simulation or for mathematical analysis of communication systems.

1.3 A general model: Marked and Weighted Point Process

A Point Process (PP) is a stochastic model used to represent the occurrence of discrete random events in some continuous domain, typically the time axis or a Cartesian plane. In general, events (and their continuous domains) can be of the most diverse nature.

In the context of source modeling, the domain is typically the time axis, and the events are connected to the generation of new traffic. Therefore, depending on the level of abstraction of the source traffic model, events can correspond to the beginning or ending of a session, or to the beginning/ending of active/inactive periods, or to the generation of new packets. In any case, we will refer to any such events as an *arrival*. Furthermore, without loss of generality, we can describe the arrival process from time zero onward.

Mathematically, the Point Process (PP) can then be represented by the set $\mathcal{T} = \{\tau_i\}$ of arrival instants, where $\tau_i \in \mathcal{R}^+$ indicates the time at which the i th arrival occurs, with $i = 1, 2, \dots$. An equivalent representation of the process can be given in terms of the so-called *inter-arrival times*, i.e., the set $\mathcal{A} = \{A_i\}$, where $A_i = \tau_i - \tau_{i-1}$ is the time between the $i-1$ st and the i th arrival. A third, equivalent way to represent the PP is by means of the associated *counting process* $N(t) : \mathcal{R} \rightarrow \mathcal{N}$, which represents the number of arrivals of the PP in the time interval $(0, t]$, i.e.,

$$N(t) = \arg \max_n \{\tau_n \leq t\} = \sum_{k=0}^{\infty} \chi \{\tau_k \leq t\} . \quad (1.1)$$

For ease of notation, the number of arrivals occurring in a given time interval $(t_1, t_2]$, with $0 \leq t_1 \leq t_2$, will be indicated as $N(t_1, t_2) = N(t_2) - N(t_1)$.

A Point Process (PP) can be extended to include additional information associated with each event, resulting in a *marked point process* or a *weighted point process*, as explained next.

1.3.1 Marked Point Process

A generalization of the PP is the Marked Point Process, where each arrival event can be associated to a certain numerosness, indicated by a compound process $\mathcal{B} = \{B_i\}$, $i = 0, 1, 2, \dots$, where $B_i \in \mathbb{N}_0$ is the *mark* associated to the i th arrival. Mathematically, the process can be represented as:

$$(\mathcal{T}, \mathcal{B}), \quad \text{or equivalently} \quad (\mathcal{A}, \mathcal{B}),$$

where B_i is the mark associated to the i th arrival. The marked point process is called *independent* if the marks B_i are independent of the arrival times τ_i . Otherwise, it is said to be *dependent*.

The marks B_i can represent characteristics such as number of Information Units (IUs) generated at the i th arrival event, the type of the IUs, or the priority. In the following, we assume that the mark actually represents the *batch size*, i.e., the number of IU generated at each arrival event.

Consider, for example, an application that generates messages to be delivered to a certain destination. Such messages can consist of multiple packets that are passed to the network layer for transmission. Then, from the perspective of the network layer, the generation of a new message by the source can be modelled as an arrival that, however, brings along a bunch of packets to be transmitted, the numerousness of which is modelled by the process $\mathcal{B}$.

1.3.2 Marked and Weighted Point Process

A Marked and Weighted Point Process (MWPP) enriches the marked point process $(\mathcal{T}, \mathcal{B})$ with the *weight* process $\mathcal{W} = \{W_{i,k}\}$, which assigns a weight $W_{i,k}$ to each IU $k \in \{1, \dots, B_i\}$ in the batch associated with arrival τ_i . The weight represents the significance or contribution of the IU for the system. For example, considering the k th IU in the i th arrival batch, $W_{i,k}$ can be the length in bits of the IU, or the energy consumption required for its transmission, or the computation required for its processing. Therefore, these weights are hence associated to a certain time of *workload*.

Therefore, a MWPP is a generalized representation of a PP that includes both marks and weights, and is hence defined by the triplet:

$$(\mathcal{T}, \mathcal{B}, \mathcal{W}) \quad \text{or equivalently} \quad (\mathcal{A}, \mathcal{B}, \mathcal{W}).$$

In this books, whenever the MWPP is used to model a packet generation process, the process $\mathcal{T}$ models the generation events, $\mathcal{B}$ indicates the batch sizes, i.e., the number of packets simultaneously generated at each arrival event,¹ and $\mathcal{W}$ represents the workload associated with each packet. In this case, the rate of the traffic generated by the application in a certain time interval $(t_1, t_2]$ can be expressed as

$$G(t_1, t_2) = \frac{\sum_{i=N(t_1)+1}^{N(t_2)} \sum_{k=1}^{B_i} W_{i,k}}{t_2 - t_1}. \quad (1.2)$$

The MWPP offers a very general model that can be specialized in many different ways to provide a variety of specific traffic source models, some of which are described in the next sections.

Example 1.3 A Consider an application that generates messages of M bytes every T seconds. This source can be represented by a MWPP with parameters $\mathcal{A} = \{T\}$, $\mathcal{B} = \{1\}$, $\mathcal{W} = \{M\}$, since arrivals occur regularly with period T , at each arrival corresponds one single message, and the message size is M .

Suppose now that each message needs to be split in K packets to be forwarded across a certain data link layer. Then, from the perspective of the data link layer, the packet generation model is a MWPP with $\mathcal{A} = \{T\}$, $\mathcal{B} = \{K\}$, $\mathcal{W} = \{M/K\}$.

1.3.3 Renewal reward processes

A specific case of a PP is the Renewal Process, which requires the inter-arrival times A_i to be independent identically distributed (i.i.d.) random variables (r.v.s) with a common

¹Typically, $B_i \geq 1$. However, it is sometimes convenient to consider $B_i \geq 0$, with $B_i = 0$ indicating events that do not bring any workload to the system. This "silent" event may correspond, e.g., to the generation of a packet that is dropped before transmission.

statistical distribution $p_{\mathcal{A}}(\cdot)$ (see Sec. ??). Each arrival represents a *renewal time* of the process, as the statistical evolution of the process no longer depends on the past, but only on the nominal distribution of the inter-arrival times.²

When the distribution of the inter-arrival times is exponential with parameter λ , then the Renewal process is a Poisson Point Process (PPP) with parameter λ (see Sec. ??). Accordingly, the number of arrivals in a time interval $(t_1, t_2]$ is a r.v. with Poisson distribution of parameters $\lambda(t_2 - t_1)$, and the arrivals in disjoint time intervals are independent.

Renewal-reward processes (see Sec. ??) extend Renewal processes by associating *rewards* $\{R_n\}$ with events. In the context of traffic modeling, the reward can be bound to the batch size, or the workload, or both. The *elementary renewal theorem* provides a foundational result for analyzing the long-term behavior of such processes. If rewards are i.i.d. with expected value $E[R]$, then the expected reward rate is given by:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \sum_{n=1}^{N(t)} R_n = \frac{E[R]}{E[A]}.$$

This theorem links the long-term average reward to the expected reward per renewal and the expected renewal time.

Example 1.3 B Consider again the Example 1.3 B. Assuming $\mathcal{B}$ and $\mathcal{W}$ are independent of $\mathcal{A}$, the average rate of the traffic generated by the source can be expressed as the ratio between the average workload generated in a renewal interval and the mean duration of a renewal interval, i.e.,

$$G = \frac{E\left[\sum_{k=1}^{B_i} W_{k,i}\right]}{E[A_j]} = \frac{E[B_\ell] E[W_{k,i}]}{E[A_j]}. \quad (1.3)$$

1.4 Classification of Source Traffic Models

The behavior of traffic generated by sources in communication networks can be classified into several distinct models, each reflecting specific characteristics of data generation and transmission. These models are critical for understanding and analyzing network performance, as they help capture the diversity of traffic patterns exhibited by different applications. In this section we list the main models, which represent a first high-level classification of traffic generation patterns. This classification provides a foundation for selecting appropriate models based on the nature of the application or network scenario under consideration.

It is important to remark that the classification of a traffic source depends on the behavior of the traffic rate that, in turn, is defined with respect to a certain time interval. By changing the reference time interval, the classification of the traffic source can also change. For example, the traffic generated by a node that sends packets of slightly variable size at regular time intervals can be seen as continuous byte stream of constant rate if observed at time intervals much larger than the inter-packet time, while at a closer look, the traffic pattern appears more variable and dynamic.

²It should be noted that this condition is generally not satisfied at time instants other than the renewal times, unless the inter-arrival times are exponentially distributed.